

Problem 1. Let $X_1, X_2, \dots, X_{80}$ be a random sample of size $n = 80$ from a distribution with probability density function

$$f(x) = \begin{cases} 6x(1-x) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(0.48 < \bar{X} < 0.52)$. Hint: Begin by calculating μ and σ^2 .

Problem 2. Suppose that a company ships packages that are variable in weight, with an average weight of 15lb and a standard deviation of 10lb. Assuming that the packages come from a large number of different customers so that it is reasonable to model their weights as independent random variables, find the probability that 100 packages will have a total weight exceeding 1700lb. (Hint: Using 68-95-99.7, calculate your final answer).

Problem 3. Let $X_1, \dots, X_n$ be a random sample from

$$f_{\theta}(x) = \begin{cases} (\theta - 1)^2 \frac{\ln(x)}{x^{\theta}} & x > 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) In this part, assume $\theta > 1$. Find the maximum likelihood estimator (MLE) for θ . (You do not need to check the 2nd derivative test)
- (b) In this part, assume $\theta > 2$. Find the method of moments estimator (MOM) for θ .

Problem 4. Let $X_1, \dots, X_n$ be a random sample from

$$f_{\theta}(x) = \begin{cases} \frac{1}{\theta} x^{\frac{1-\theta}{\theta}} & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the maximum likelihood estimator (MLE) for θ . (You do not need to check the 2nd derivative test)
- (b) Find the method of moments estimator (MOM) for θ .

Problem 5. Consider two continuous random variables X and Y with joint p.d.f.

$$f_{X,Y}(x,y) = \begin{cases} Ce^{-(x+2y)} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (5-1) Find the value of C so that $f_{X,Y}(x,y)$ is a valid joint p.d.f.
- (5-2) Find the marginal probability density function of X , $f_X(x)$. What type of distribution X has?
- (5-3) $E[3X(X - Y)]$
- (5-4) Are X and Y independent?
- (5-5) Calculate $E[XY]$ using table of distributions.
- (5-6) Find the conditional density of X given that $Y = y$ for some $y > 0$.
- (5-7) Evaluate $\text{Var}(-2X+2)$
- (5-8) Evaluate $E[3X - 2|Y]$
- (5-9) Evaluate $E[(X - 2Y)^2]$

Problem 6. A book has n typos. Two proofreaders, Sarah and Hannah, independently read the book. Sarah catches each typo with probability p_1 and misses it with probability $1 - p_1$, independently, and likewise for Hannah, who has probabilities p_2 of catching and $1 - p_2$ of missing each typo. Let X_1 be the number of typos caught by Sarah, X_2 be the number caught by Hannah, and X be the number caught by at least one of the two proofreaders.

- (a) Find the distribution of X .
- (b) For this part only, assume that $p_1 = p_2$. Find the conditional distribution of X_1 given that $X_1 + X_2 = t$

Problem 7. Let X and Y be two independent random variables with the CDFs F_X and F_Y . Define

$$Z = \max(X, 2Y), \\ W = \min(X, Y)$$

Find the CDFs of Z and W .

Problem 8. A fair 4-sided die is tossed twice, independently. Let the random variable Z be defined as the number of dots seen on the first toss minus the number of dots seen on the second toss. Find the expected value of Z^2 .

Problem 9.

$x = 1$	$\frac{1}{6}$	$\frac{1}{8}$	$\frac{1}{4}$
$x = 0$	$\frac{1}{12}$	$\frac{1}{8}$	$\frac{1}{4}$
	$y = -1$	$y = 0$	$y = 1$

- (a) Find $P(X = 0|Y^2 = 1)$.
- (b) Find the marginal PMFs of Y .
- (c) Find $P(X = 1|Y = 0)$.

Problem 10. Suppose that a lecture note with 20 pages contains on average 4 misprints per page. What is the probability that there will be at least 8 pages which contain more than 5 misprints?

Problem 11. Consider 10 independent tosses of a biased coin with the probability of Heads at each toss equal to p , where $0 < p < 1$.

- (a) Let A be the event that there are 6 Heads in the first 8 tosses. Let B be the event that the 9th toss results in Heads. Find $P(B|A)$.
- (b) Find the probability that there are 3 Heads in the first 4 tosses and 2 Heads in the last 3 tosses.
- (c) Given that there were 4 Heads in the first 7 tosses, find the probability that the 2nd Heads occurred at the 4th toss.
- (d) Calculate the probability that there are 5 Heads in the first 6 tosses and 3 Heads in the last 5 tosses.

Problem 12. During one day Sarah receives X legitimate emails and Y spam emails. Suppose X and Y are independent, $X \sim \text{Pois}(\lambda_x)$ and $Y \sim \text{Pois}(\lambda_y)$. Assuming one day Sarah receives n emails. Given this information what is the expected value of how many legitimate email she has? Show all the work.

Problem 13. Let $X_1, X_2, \dots, X_{300}$ be a random sample of size $n = 300$ from a distribution with probability density function

$$f(x) = \begin{cases} \frac{3}{x^4} & x > 1 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(1.6 < \bar{X}_n < 1.65)$ using central limit theorem.
Hint: Begin by calculating μ and σ^2 .
Hint: If needed at the end use 68-95-99.7 Rule.

Problem 14. Let $X_1, X_2, \dots, X_n$ be I.I.D exponential random variables with parameter λ . Let $Y_1, Y_2, \dots, Y_n$ be I.I.D Bernoulli random variables with parameter $p_i = P(X_i \geq 1)$. Answer to the following questions:

- (a) what is distribution of $Y_1 + Y_2 + \dots + Y_n$ when n is small? Write down the name of distribution with its parameter/s.
- (b) what is distribution of $Y_1 + Y_2 + \dots + Y_n$ when n is large? Write down the name of distribution with its parameter/s.
- (c) Let $S_n = Y_1 + Y_2 + \dots + Y_n$. Assume $n=200$ and $\lambda = \ln(3)$. What is the $P(S_n > 50)$

Problem 15. It is estimated that 50% of emails are spam emails. Some software has been applied to filter these spam emails before they reach your inbox. A certain brand of software claims that it can detect 99% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 5%.

Now if an email is detected as spam, then what is the probability that it is in fact a non-spam email?